

A Self-Closing Gate Conundrum

#1 Victory Royale

Element D: Design Concept Generation, Analysis, and Selection

Student G's family needs a self-closing mechanism that closes their family's gate on its own because his family does not want unwanted animals and people to get inside their fence gate because it is a safety concern that could cause injury. This happens because student G's gate can not close on its own.

Our group brainstormed what would close the gate, we came up with as many ideas as possible. Then narrowed down our ideas to the top five that fit our criteria. Through our decision matrix we found the solution that best fit our goals.

Our group's final decision was a tension spring. We chose this solution because it fit our design requirements the best. Our design was more durable, inexpensive, and compact. It also was easier to install than any other design.

We will build our solution by taking measurements of the fence and gate. Our final solution consists of a tension spring that will be mounted to the fence and gate, automatically closing it. Our estimated materials cost was \$45. Our materials list consisted of a tension spring, metal strap, bolts, and nuts.

- 1 Spring closing gate
- 2 Hydraulic closing gate
- 3 Weight and pulley
- 4 Compression Spring
- 5 mechanical wheel

- Spring closing gate. ☆
- Hydraulic closing gate. ☆
- String closing gate. X
- Weight and pulley system. ☆
- ~~Pullable~~ Pullable lever that closes the gate. X
- Close it with self. X
- Climb the gate. X
- Closes when it reaches a certain time. X
- Foot pedal that closes the gate. ☆
- No gate. X
- Magnetic gate. X
- Wheeled gate. ☆